

Pete's Hornet Simulator Matrix Reliability Update - Revision 3

Technical handover for the four revised Mega 2560 matrix controller sketches - Revision 3

Purpose. This document records the matrix-scanning and input-qualification changes applied to Pete's RHS, LHS, LIP and UIP controllers. It is aimed at a competent developer and deliberately describes the implementation, design trade-offs, validation performed, and outstanding behavioural caveats in detail.

Scope: matrix scan sequencing, physical input acquisition, candidate-state qualification, startup state capture and selective DCS-BIOS resend behaviour. Controller-specific button/switch mappings were intentionally not changed in the original four-sketch matrix pass. Revision 2 intentionally removes AMPCD, UFC and IFEI command paths from Pete's UIP only. Revision 3 adds a bounded AOA indexer brightness path on Pete's UIP: the existing A6 pot is retained, but its DCS output is clamped to 75%-100% rather than allowing a zero output.

Important: all descriptions below reflect the delivered honed sketches as generated. This is a technical record, not a claim that every possible electrical-noise source has been eliminated. Software qualification rejects transient readings; persistent electrical faults still require hardware investigation.

1. Source Files, Scope Control and Verification

The work was constrained to preserve Pete's established simulator architecture. Networking, reflector/debug transport, MSFS pathways, hardware outputs, stepper/servo behaviour, backlighting and startup logic were left in place unless the matrix implementation required a local integration point. In Revision 2, the UIP AMPCD/UFC/IFEI DCS-BIOS paths were deliberately retired because those controls are no longer assigned to that controller.

Controller	Original	Revised	SHA-256 original	SHA-256 revised
RHS	RIGHT_CONSOLE_COMBINED_CONTROLLER.ino	Petes_RHS_matrix_honed.ino	89422f726e295...f45c6a	c3167d994efe...4f9042
LHS	LEFT_CONSOLE_INPUT_CONTROLLER_V2.ino	Petes_LHS_matrix_honed.ino	d884ca0977fa...20aee	c69f537eee96...bb37b
LIP	LIP_CONTROLLER.ino	Petes_LIP_matrix_honed.ino	14f0855b366e...f8ed9	158e5eaed880...ed8a2
UIP	UIP_COMBINED_CONTROLLER.ino	Petes_UIP_matrix_honed.ino	1d964ca649c8...95dea	829feb789c62...4fef3

Verification performed: after normalising line endings, the full text of ``void CreateDcsBiosMessage(int ind, int state)`` was extracted from each original/revised pair and compared. RHS, LHS and LIP matched exactly. UIP matched exactly at the end of the original matrix pass. In Revision 2, the UIP function intentionally differs only because AMPCD, UFC and IFEI command statements are disabled; a further command-level comparison confirmed that there are no removals or additions outside those three groups.

Verification boundary. The comparison establishes mapping-function identity, not runtime certification. These sketches were reviewed statically and documented, but were not compiled/uploaded against Pete's exact local Arduino/DCS-BIOS library versions or hardware in this environment.

2. Common Hardware/Matrix Topology

All four controllers use the Mega 2560 in the same logical matrix topology: sixteen active-low rows and eleven active-low columns. Each physical matrix position is addressed by ``inputIndex = row * 11 + column``, yielding 176 physical positions numbered 0 to 175.

Matrix element	Implementation
Rows 0-7	Mega D22-D29 / PORTA bits 0-7; one bit is driven low for the selected row.
Rows 8-15	Mega D30-D37 / PORTC bits 7-0; one bit is driven low for the selected row.
Columns 0-3	D38 PD7; D39 PG2; D40 PG1; D41 PG0.
Columns 4-10	D42-D48 / PORTL bits 7 through 1.
Not used as matrix columns	D49 intentionally unused; D50-D53 reserved/unavailable because of SPI/Ethernet/SD use.
Electrical convention	Columns use INPUT_PULLUP. A closed switch on the selected active-low row reads low and is stored as logical state 1.

3. Original Matrix Issue and Why the Scan Was Rewritten

The original scan in each sketch used a 16-element ``colResult[]`` and looped ``colid`` from 0 to 15, even though only eleven physical columns were wired/read. Columns 11-15 were hard-coded to inactive. The state-array index was nevertheless calculated with an eleven-column stride:

```
joyReport.button[(rowid * 11) + colid] = ...;
```

This combined a 16-column iteration with an 11-column address calculation. The additional five virtual columns therefore addressed non-physical positions and, for rows 0-14, temporarily pointed into the beginning of the next logical row. Because the next row was later scanned, most of those temporary writes were corrected before input processing; however, the code was structurally inconsistent, performed unnecessary state writes, processed non-physical indices and made fault isolation unnecessarily difficult. On the final row it also wrote beyond the intended 176 physical-position domain, albeit still within the 256-element software array.

Every revised sketch now uses a strict 16 x 11 scan and writes only indices 0-175. The column read map is explicit and fixed to the eleven real pins. This removes the logical mismatch without changing the physical wiring or the existing input numbering.

4. Common Reliability Pattern

The revised software is endpoint-qualified rather than event-triggered. A raw transition does not immediately issue a DCS command. It first becomes a candidate state; only if the same raw state remains present for the required stability interval is it committed as the new confirmed state.

```
raw change -> candidate state / candidate timestamp
raw remains unchanged for required interval -> confirmed state -> normal command path
raw returns to previous confirmed state before timeout -> discard candidate, no command
```

This is materially different from the original lockout-style implementation. The old approach could send on the first observed edge, then ignore follow-up movement during the debounce period. The revised approach waits for a stable endpoint before it sends anything. That is more appropriate for matrix wiring exposed to contact bounce, row/column settling and short-duration PWM-coupled noise.

Stage	Revised behaviour
Row selection	Both row banks are released (`PORTA = 0xFF`, `PORTC = 0xFF`) before the next row is driven low.
Electrical settle	The existing 80 us settle period is retained after row selection. At the base level this adds $16 \times 80 \text{ us} = 1.28 \text{ ms}$ per complete matrix pass, excluding port reads and application work.
Acquisition	Only eleven valid columns are sampled; each is written to `row * 11 + column`.
Startup	The first completed scan copies current physical states into the confirmed-state array and emits no normal input command.
Transient rejection	Any candidate that changes again or returns to the confirmed state before timeout is discarded.
DCS resend	Where enabled for maintained controls, the confirmed value is transmitted again after 100 ms, provided the confirmed state has not changed again.

Startup consequence. The deliberate no-command startup policy prevents a powered-up panel from moving DCS controls. The trade-off is that the physical panel and DCS may begin mismatched after a reset; each control will synchronise when it is physically moved, unless another explicit state-synchronisation mechanism is used.

5. DCS-Only Confirmation Resend

For maintained controls, a second transmission of the confirmed DCS value is scheduled 100 ms after the accepted change. The resend is cancelled if the confirmed state has changed in the meantime. The aim is to make a rare first-command delivery failure non-material while keeping the input logic deterministic.

The resend is appropriate for a true maintained switch because sending the same target state twice is normally idempotent. It must not be applied indiscriminately to momentary controls, because a second asserted state can be semantically different from a held or repeated button action. This is why the LIP and UIP revisions add input-type policy. The RHS and LHS revisions do not yet distinguish input types; see the controller-specific caveats.

6. RHS - Right Console

Delivered file: `Petes\_RHS\_matrix\_honed.ino`

Setting	Value / behaviour
Physical matrix	16 rows x 11 columns; 176 physical inputs.
Row settling	80 us (retained from original).
Stable-state qualification	50 ms for every matrix position.
Startup policy	First full scan is captured only; no DCS/MSFS/debug commands are intentionally issued from the matrix state capture.
Confirmation resend	Enabled for every matrix position; 100 ms after the confirmed command.
MSFS behaviour	Normal MSFS state message is sent only on the initial confirmed transition. The scheduled resend does not call `SendMSFSMessage()`.
Debug/reflector behaviour	See important note below: the resend enters the unchanged mapping helper and can emit command-level debug traffic.

Specific code changes:

- Introduced `MATRIX\_ROWS`, `MATRIX\_COLUMNS`, `MATRIX\_INPUTS`, `ScanDelay`, `DebounceDelay` and `DCS\_RESEND\_DELAY` constants.

- Changed `bFirstTime` from false to true so the first complete scan establishes the baseline rather than treating initial physical positions as changes.
- Added `dcsResendPending[]`, `dcsResendState[]` and `dcsResendAt[]` for per-input resend scheduling.
- Replaced the 16-column scan with an explicit eleven-column `const byte colResult[MATRIX_COLUMNS]` map.
- Before each row selection, both PORTA and PORTC are returned to `0xFF`, then the relevant active-low row is selected.

RHS caveat. The RHS revision treats all 176 inputs as maintained controls. This was a conservative carry-over of the initial side-console strategy, but it means any physical momentary button on the RHS is also subject to 50 ms qualification and the 100 ms DCS resend. A quick press that has already been released before the scheduled resend will cancel the resend; a press held longer than 100 ms may be re-transmitted. If the RHS contains momentary functions that are sensitive to repeat assertion, add an explicit momentary-input policy before relying on this as the permanent production version.

RHS debug nuance. The outer matrix resend does not call `SendMSFSMessage()` or its matrix-level debug line. However, `CreateDcsBiosMessage()` calls the unchanged `sendToDcsBiosMessage()` helper, and that helper calls `SendDebug()`. Therefore the DCS resend can generate duplicate command-level reflector/debug messages. It is harmless to DCS operation, but it is not completely silent at the debug-transport layer.

7. LHS - Left Console

Delivered file: `Petes_LHS_matrix_honed.ino`

Setting	Value / behaviour
Physical matrix	16 rows x 11 columns; 176 physical inputs.
Row settling	80 us (retained).
Stable-state qualification	50 ms for every matrix position.
Startup policy	First full scan is captured only; no matrix-originated DCS/IP/debug actions are sent.
Confirmation resend	Enabled for every matrix position; 100 ms after confirmed state.
Network/debug	Core Ethernet and reflector code was preserved. See debug caveat below.
Other subsystems	Existing exterior/interior-light logic, APU/engine magnetic switch logic, rudder trim, brake-pressure stepper and startup sequencing remain outside the matrix rewrite.

The LHS code changes are structurally the same as RHS: strict 16 x 11 scanning, explicit row-bank reset, 80 us settling, baseline-only first scan, 50 ms endpoint qualification and 100 ms DCS confirmation resend.

LHS caveat. As delivered, LHS also applies maintained-switch behaviour to every matrix position. Its existing mapping includes both maintained and button-like controls. The change preserves mappings, but not every physical control has a momentary exemption. Any function that should not see a repeated asserted command should be added to a specific momentary list, following the LIP/UIP pattern.

LHS debug nuance. The scheduled resend reaches the unchanged `sendToDcsBiosMessage()` helper, which itself sends a reflector/debug message. Thus the scheduled DCS resend may create duplicate low-level debug traffic. The main matrix-level debug event and non-DCS integrations are not intentionally repeated, but command-level debug is not suppressed in this revision.

8. LIP - Lower Instrument Panel

Delivered file: `Petes_LIP_matrix_honed.ino`

Setting	Value / behaviour
Physical matrix	16 rows x 11 columns; 176 physical inputs.
Row settling	80 us.
Momentary qualification	20 ms.
Maintained qualification	50 ms.
Startup policy	<code>SEND_INITIAL_STATES = false</code> ; physical positions are learned but not sent on boot.
Maintained resend	Enabled; DCS-only 100 ms confirmation resend.
Momentary resend	Disabled.
Debug on resend	Suppressed through <code>suppressMatrixResendDebug</code> ; normal debug remains intact.

Unlike RHS/LHS, LIP is input-type aware. It introduces `candidateButton[]`, `candidateChangedAt[]`, `resendPending[]`, `resendState[]` and `resendDueAt[]`. Raw input movement starts/restarts the candidate timer. A normal command only occurs after the candidate is stable for the relevant interval.

8.1 LIP Inputs Explicitly Treated as Momentary

The following physical matrix positions use 20 ms confirmation and never receive the 100 ms resend:

Group	Matrix positions	Mapped functions (as named in existing code)
Select Jett	17, 28, 39, 50, 61	SJ_CTR, SJ_LI, SJ_RI, SJ_RO, SJ_LO. Existing press/release command mapping and latching behaviour retained.
AMPCD PBs	77-80, 88-91, 99-102, 110-113, 121-124	AMPCD_PB_01 through AMPCD_PB_20 in the existing non-linear physical layout.
IFEI buttons	143, 144, 154, 155, 165, 166	IFEI_MODE_BTN, IFEI_DWN_BTN, IFEI_QTY_BTN, IFEI_ZONE_BTN, IFEI_UP_BTN, IFEI_ET_BTN.

All other LIP inputs default to the maintained-switch policy: 50 ms stable state plus one DCS-only resend after 100 ms. This reflects the explicit set of button groups requested for the LIP update. It is intentionally a narrow classification, not a claim that no other LIP input is momentary.

LIP follow-up point. If other LIP controls later prove to be physical momentary buttons - for example a discrete RWR, CMSD or auxiliary button - add their matrix index to `IsMomentaryInput()`. No assignment change is required; it only changes qualification/resend policy.

The LIP resend calls `CreateDcsBiosMessage()` with `suppressMatrixResendDebug = true`. That retains normal debug on the first confirmed transition but intentionally prevents the scheduled DCS resend from generating duplicate reflector/debug output. The resend also does not call `SendMSFSMessage()` again.

9. UIP - Upper Instrument Panel

Delivered file: `Petes_UIP_matrix_honed_v4_no_AMPCD_UFC_IFEI_AOA_75min.ino`

Setting	Value / behaviour
Physical matrix	16 rows x 11 columns; 176 physical inputs.
Row settling	80 us.
Default classification	Momentary: 20 ms stable state and no resend.
Known maintained classification	50 ms stable state and one DCS-only resend after 100 ms.
Startup policy	<code>SEND_INITIAL_MATRIX_STATES = false</code> ; baseline only, no boot command burst.
Debug on resend	Suppressed via <code>suppressMatrixResendDebug</code> .
Stepper scheduling	<code>updateSteppers()</code> remains inside the scan workflow and after input processing, preserving service during matrix activity.

The UIP is treated differently because it is predominantly push buttons. The practical runtime rule is: all active inputs default to momentary behaviour unless their index is explicitly returned by `IsMaintainedInput()`. `StableTimeForInput()` is driven by this maintained list. Revision 2 adds `IsDisabledInput()` ahead of the normal qualification/dispatch path so the retired AMPCD/UFC/IFEI matrix addresses are still scanned but remain application-layer silent.

9.1 UIP Momentary Groups

- Left and right DDI push-button groups, including the validated positions for left fire / master caution and right fire / APU fire.
- AMPCD is no longer active on this UIP revision. Its original physical positions remain in the scan map but are explicitly disabled by `IsDisabledInput()`.
- UFC and IFEI are no longer active on this UIP revision. Their original physical positions remain in the scan map but are explicitly disabled by `IsDisabledInput()`.
- Select Jett positions 171-175, explicitly treated as push buttons.

The document does not reproduce all momentary indices in prose because the actual authoritative list is in `IsMomentaryInput()` and remains linked to the existing validated `CreateDcsBiosMessage()` mapping. The source code includes the full indexed list for code review.

9.2 UIP Maintained Inputs

Only active indices in `IsMaintainedInput()` receive the 50 ms / 100 ms maintained policy. Retired AMPCD positions are excluded from this policy. The active maintained groups cover:

- DDI brightness/contrast and directional selector positions: 4, 9, 10, 15, 20, 21, 26, 31.
- ECM selector positions: 85, 96, 107, 108, 118, 126, 129, 137, 148, 159 and 170. The AMPCD-specific positions 110-113, 121-124 and 132 are now retired/inert.
- HUD, master-arm, mode and other maintained controls: 139-142, 146, 150-153, 156-157, 161-162, 164 and 167-168.

The UIP uses helper functions ``ReadMatrixColumn()``, ``AllMatrixRowsOff()``, ``SelectMatrixRow()`` and ``ScanMatrix()`` rather than retaining the long inline scan. This improves auditability without changing row/column wiring or input numbering.

10. What Was Intentionally Not Changed

- No ``CreateDcsBiosMessage()`` mappings, values, switch positions, fire/master-caution positions, Select Jett assignments, cover timing logic or existing button latching were altered in RHS, LHS or LIP. In UIP Revision 2 only, AMPCD/UFC/IFEI command paths are intentionally disabled; all other UIP mappings remain unchanged.
- No Ethernet addressing, MAC/IP configuration, reflector target, keepalive protocol, debug message format, MSFS message mapping, external network protocol or DCS-BIOS transport setting was redesigned.
- No change was made to panel lighting, backlight calibration, MAX7219 wiring, stepper motor control, magnet-switch logic or output buffer addresses as part of this Pete-specific pass. The sole analogue-control exception is the Revision 3 UIP AOA indexer brightness input on A6, which remains physically connected but is now software-clamped to 75%-100% before sending the DCS-BIOS HUD_AOA_INDEXER command.
- No electrical hardware modification was made. The revised code filters short-duration false readings but does not replace proper grounding, pull-ups, decoupling, cable separation, diode isolation or shielding where the underlying disturbance is persistent.

11. Recommended Validation Procedure

- Upload one controller at a time and preserve a copy of its prior known-good binary/source state.
- With DCS disconnected or aircraft paused, power-cycle the controller and confirm that the first scan does not generate an input burst. Debug reflector logs should show setup activity only, not 176 control transitions.
- For each maintained switch, move it through every physical state. Confirm one normal DCS transition and, where configured, the harmless confirmation resend. Verify no false transitions while stationary.
- For each momentary group on LIP/UIP, make quick taps shorter than 100 ms and normal longer presses. Confirm a single press/release semantic and no scheduled second press.
- Stress the matrix while PWM/backlighting is active, dimming, or carrying the noisiest load. Confirm that a short false low does not survive the 20 ms or 50 ms qualification interval.
- For RHS/LHS, specifically test any physical momentary functions. If a momentary command is repeat-sensitive, add a momentary exemption rather than disabling the entire reliability framework.
- If debug/reflector traffic volume matters, observe RHS/LHS during a maintained-switch transition. Their DCS confirmation resend can generate duplicated command-level debug messages; LIP/UIP intentionally suppress those duplicate logs.

12. Failure Interpretation / Diagnostic Notes

Observed behaviour	Most likely layer	Suggested response
False event that disappears within a few milliseconds	Matrix noise/contact bounce	Revised software should reject it. If not, check actual duration against 20/50 ms settings.
Repeated false event after >50 ms	Persistent electrical coupling, weak pull-up, shared return or wiring fault	Investigate hardware; software is correctly seeing a stable false state.
Only one row/column combination misbehaves	Local switch, diode, ribbon conductor, row/column trace or connector	Swap known-good switch/wiring position and compare whether fault follows hardware or address.
DCS action missing while debug shows confirmed command	DCS-BIOS transport/module/control mapping	The 100 ms resend should reduce this for maintained controls; inspect DCS-BIOS and serial/IRQ path.
DCS action duplicated on a button	Momentary input incorrectly classified as maintained	Add that matrix index to the momentary group; do not alter its mapping.
DCS and physical panel mismatched after boot	Intentional baseline-only startup policy	Move the control or implement a deliberate boot synchronisation layer if desired.

13. Final Technical Position

The central matrix defect has been removed: each controller now reads the physical 16 x 11 topology directly, with explicit row isolation and a fixed settling interval. The input path is no longer driven by

the first observed edge; it now commits only stable end states. LIP and UIP add the more appropriate distinction between momentary and maintained controls. Revision 2 leaves the matrix architecture intact while retiring the now-unassigned AMPCD/UFC/IFEI UIP command paths. Revision 3 additionally prevents the UIP AOA brightness input from commanding a value below 75%, removing a known path by which HUD_AOA_INDEXER could be dimmed to zero.

The deliberate design decision is that matrix integrity is now handled at three layers: electrical row settling, software state qualification, and selective DCS confirmation resend. This should materially reduce intermittent missed or false commands caused by bounce, scan timing and transient PWM/LED coupling. It is not a substitute for correcting a line that is persistently noisy for longer than the qualification interval.

Key review item for Pete. The two side consoles are intentionally simpler but less type-aware: their resend policy applies to all matrix positions and their scheduled DCS resend can still emit command-level debug traffic. The LIP/UIP pattern is the reference implementation for any future refinement: explicit momentary/maintained policy, one normal event, one DCS-only maintained resend, and debug suppression for the resend.

14. Revision 2 - UIP AMPCD / UFC / IFEI Removal

Purpose. AMPCD, UFC and IFEI are no longer required on Pete's UIP. Their physical inputs are retained in the 16 x 11 scan topology so that every remaining matrix address, row/column assignment and existing wiring reference remains unchanged. The retired positions are no longer allowed to dispatch DCS-BIOS controls, generate matrix debug/reflector traffic, or schedule a confirmation resend.

Implementation. The revised UIP adds `IsDisabledInput(int ind)`. In `FindInputChanges()`, disabled addresses are synchronised directly to `prevjoyReport` and `candidateButton`, their resend flag is cleared, and the loop continues before `CreateDcsBiosMessage()` or `SendDebug()` can run. The command lines for the retired AMPCD/UFC/IFEI controls inside `CreateDcsBiosMessage()` are additionally commented as `UIP removed` for source-level clarity. This creates a deliberate double guard: the dispatcher cannot call those paths in normal operation, and those case branches contain no active DCS-BIOS send statements.

No analogue declarations were removed for the Revision 2 AMPCD/UFC/IFEI scope reduction. Pete's UIP did not contain dedicated AMPCD, UFC or IFEI `DcsBios::Potentiometer...` declarations. AMPCD brightness on this controller was represented by a matrix command path, not by a separate analogue object. In Revision 3, only the existing HUD AOA indexer declaration is replaced by a bounded manual A6 handler; all other LDDI/RDDI, HMD and HUD analogue declarations remain unchanged.

14.1 Retired Physical Matrix Addresses

Group	Retired matrix indices	Removed DCS-BIOS command paths
AMPCD	55-58, 66-69, 77-80, 88-91, 99-102, 110-113, 121-124, 132	PB 1-20; GAIN, CONT, SYM, NIGHT/DAY and BRT controls.
UFC	32, 37, 48, 59-65, 70-76, 81-84, 92-95, 103-106, 114	ILS, COMM pulls, keypad 0-9, CLR, ENT, OS1-OS5, AP, IFF, TCN, ON/OFF, BCN, DL, ADF, IP and EMCON.
IFEI	143, 144, 154, 155, 165, 166	MODE, DWN, QTY, ZONE, UP and ET.

14.2 Verification of the Scope Reduction

- The original UIP `CreateDcsBiosMessage()` function contained 319 active DCS-BIOS send statements. The Revision 2 UIP contains 189 active send statements. The 130 disabled statements are exclusively AMPCD, UFC or IFEI paths; no non-AMPCD/UFC/IFEI DCS-BIOS send statement was removed or added.
- The command-level comparison was performed after extracting the same function from the prior UIP honed sketch and Revision 2. All removals were verified to have a control name beginning `AMPCD_`, `UFC_` or `IFEI_`. The revised source contains no active `sendToDcsBiosMessage()` statement for those three groups.
- The physical matrix still scans all 176 inputs. Retired positions are deliberately not removed from the scan because omitting them would change the row * 11 + column indexing model and risk moving unrelated validated inputs. They are simply silent at the application layer.
- Practical test: press any retired AMPCD/UFC/IFEI control and confirm that DCS receives no command and the reflector/debug output does not show a `Front Input` transition for that position. Then check a neighbouring active control to confirm its original mapping and stable-state behaviour are unaffected.

15. Revision 3 - UIP AOA Indexer Brightness Clamp

Purpose. The original UIP used `DcsBios::PotentiometerEWMA<5, 512, 10>`

`hudAoaIndexer("HUD_AOA_INDEXER", A6)`. That permitted the A6 potentiometer to transmit the full 0-65535 DCS-BIOS range. A low, disconnected, noisy or briefly invalid A6 reading could therefore command HUD_AOA_INDEXER to zero. This is relevant beyond the physical MAX7219/LED output: DCS can also blank the in-game/video AOA presentation when the indexer brightness is commanded too low.

Implementation. The stock hudAoaIndexer PotentiometerEWMA declaration is commented out and replaced with a dedicated manual handler. analogRead(A6) retains the original non-inverted pot direction but maps 0-1023 to 49152-65535, corresponding to 75%-100% of the 16-bit DCS-BIOS control range. The mapped value is explicitly clamped at both ends before it is transmitted as HUD_AOA_INDEXER. Thus the pot's minimum position is now 75%, not 0%, and its maximum remains 100%.

Timing and transport. The handler is invoked once immediately after DcsBios::setup() and then at a 100 ms cadence from the main loop. It sends through sendDcsBiosMessage() directly, conditional on DCSBIOS_In_Use, rather than through the matrix sendToDcsBiosMessage() wrapper. This intentionally avoids reflector/debug messages for a periodic analogue brightness update and leaves all matrix event/debug semantics unchanged.

Scope boundary. No AOA LED buffer addresses, MAX7219 positions, AOA high/normal/low output handlers, HUD stepper code, button mappings, AMPCD/UFC/IFEI scope reduction, Ethernet configuration or reflector protocol is changed by this revision. The only functional change is the replacement of the unrestricted A6 brightness sender with a bounded 75%-100% sender.

Recommended test. Start with the A6 pot fully counter-clockwise and verify that the in-game/video AOA indicator no longer blanks. Move the pot through its travel and confirm a continuous brightness change over the remaining 75%-100% range. Then test the physical AOA indexer in an approach/landing state and during the pilot light-test sequence. If the in-game/video AOA remains absent while HUD_AOA_INDEXER is confirmed above 49152, the next diagnostic focus should be aircraft/DCS state or the DCS-BIOS output bits rather than the A6 brightness path.